

IBM Employee Retention Report

Identifying Employee Retention through People Analytics

Prepared for

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July 2026

Application link: <https://5d63c594-b5d8-4808-ac8d-b2217e7fbc4a.plotly.app>

Problem

Dr Sarah Chen is concerned about the employee retention rate at IBM. As CHRO (Chief Human Resources Officer) Dr Chen realizes that there is a growing cost when it comes to replacing mid level employees which puts strain on the employee pipeline. Efforts have been made to understand why employees are exiting through exit interviews and engagement surveys. However, the HR team has still struggled to identify which employees are at risk of exiting and the current set up lacks data driven approach to further understand the reason why employees are exiting. Dr Chen knows that the data gathered from the employees' profile can tell us who is at risk of leaving and it is our job as the consulting team to develop an application that will uncover natural employee segments and which segment is at a higher risk for exiting.

Data

Data provided via csv file containing 1,470 records. Data provided from column left to right, Age, Attrition, Department, Education, Gender, JobLevel, JobSatisfaction, MonthlyIncome, TotalWorkingYears, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsWithCurrManager.

Insights

Using the 13 variables given from the data set we applied the k-means clustering. Using the elbow method and silhouette score we determined that $k = 4$ is the optimal number of clusters producing our four segments. Then, we profiled them by job level, tenure, and demographics to find patterns in attrition.

Deployment

An interactive app was deployed using Cloud Plotly using the dash framework containing cluster quality and K mean visualization, cluster segment exploration, and an employee scenario simulator. Python is the programming language being used.

Vibe-Coding Prompt to Generate App

"You are a data analytics consultant for IBM instructed by the Chief Human Resources Officer to investigate employee retention. See the attached data set with the following columns (data set attached and columns described). Provide a python script that performs the following.

Set attrition aside to check clusters against. Keep the ordinal variables as is (variables listed) Gender needs to be a dummy variable (0 or 1) indicating male or female. A one-hot encoded department and standardize all features so no single variable dominates distance.

Apply the k-means clustering. Compute and display both the elbow method and silhouette score.

For the chosen k, create a profile for each cluster label that captures the business meaning of the segment ("High-Risk Early Career", "Loyalty Anchor", "Mid-Career Stable"). Profile each label by employee count, attrition rate, average job level, income, tenure, satisfaction, work-life balance, and department mix

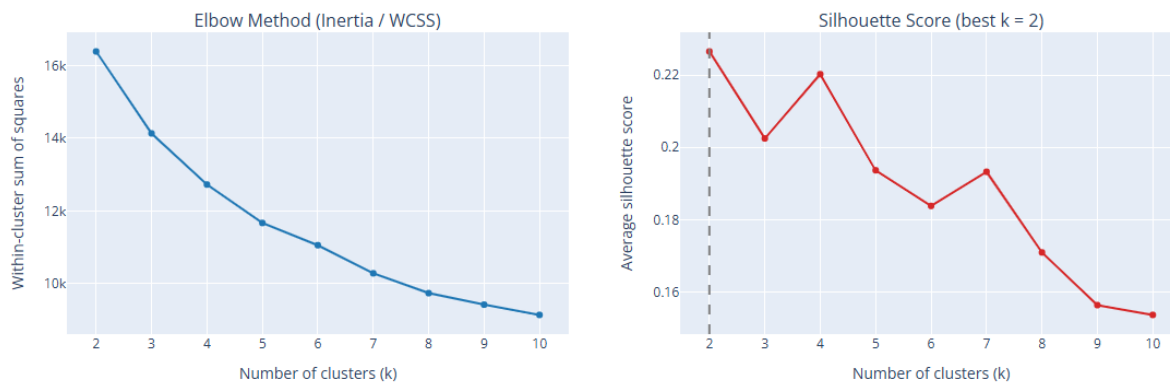
Build a scenario simulator that takes inputs of the employee's characteristics and assigns them to the nearest existing cluster and estimates the hypothetical employees' profile and attrition rate. Build the

application interface using the dash framework structure to then be deployed as a public plotly cloud application. Also include cluster visualizations, profile comparison charts, and KPIs to go along with the application to help technical and non-technical audiences visualize.

Pre-processing & Validation

We start with setting the attrition aside as we are checking against this. Variables containing integers were checked and left as is. Gender has been set as a dummy variable (0 or 1), department has been one-hot encoded, and used standard scaler so no single variable dominates distance.

Elbow Method and Silhouette Score

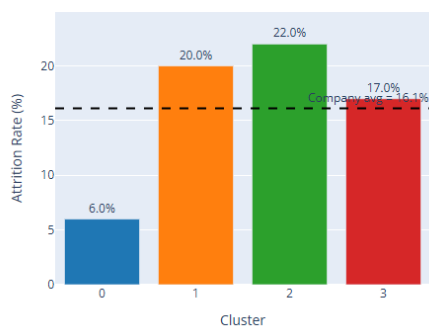


The elbow method is used to determine an optimal number of clusters. The plot is highest at k=2 and bends through k 3-5 with gains diminishing gradually after that. For the silhouette score is used to evaluate the quality of clustering measuring how well each data point is assigned to its cluster (formula below). The silhouette peaks at K = 2 (0.227) and second at K = 4 (0.22) then declines from there.

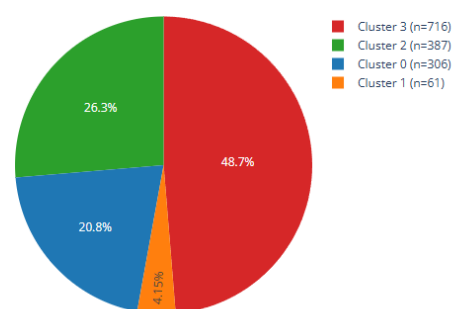
Cluster Profile evaluation

Solution fitted with k = 4 clusters | Silhouette Score = 0.2202 | Inertia = 12725

Attrition Rate by Segment



Workforce Share by Segment



Cluster	Employees	AttritionRate	AvgAge	AvgJobLevel	AvgIncome	AvgTotalWorkingYears	AvgYearsAtCompany	AvgYearsInRole	AvgYearsWithManager	AvgWorkLifeBalance
0	306	6	45.07	3.55	12988.45	22.28	14.24	7.76	7.43	2.77
1	61	20	37.39	1.95	6344.39	10.87	6.41	3.43	3.51	2.92
2	387	22	34.89	2.01	5831.88	8.87	5.7	3.88	3.62	2.82
3	716	17	34.5	1.47	4107.4	7.91	4.67	2.97	3.03	2.71

Cluster 0 (Senior) contains the senior, highest paid, and longest tenure (13k a month, 14.2-year tenure). This mainly contains the R&D team with few sales presence. This cluster signals very low risk in employees exiting as senior status, high pay and high tenure are all strong predictors. This will be a low-risk cluster, however possible retirement may be possible for such high tenured employees.

Cluster 1 (HR) while small the attrition rate is at 20% well above the 16.1% attrition rate average. This cluster comprises of all HR team members with tenures at 6.4 years containing the highest work life balance at 2.92. This high attrition could be a possible limitation in career mobility and pathing may be worth investigating.

Cluster 2 (Early Career Sales) contains the highest risk with this segment containing young, low tenure, and lowest pay sales staff. The average age is about 35 years, tenure averaging 5.7 years, and paying 5.8k a month. With an attrition rate of 22%, this is a cluster that needs the most attention with recommendations to invest in onboarding, workload management, and expectations.

Cluster 3 (Early career R&D) contains the young, lowest pay, and shortest tenure employees, containing most of the staff. This cluster contains a 4.7-year tenure, less than 4k a month pay with a attrition of 17% being near the company average. While one of the highest risks, this is 49% of the staff specifically R&D. An investigation into onboarding and expectations would be a good start.

Questions

What characterizes employees who leave IBM compared to those who remain?

What characterizes employees who stay are mostly in cluster 0 with higher pay, tenure, older, and level of seniority contributing to someone who stays. What characterizes someone as an exit risk are young, lower income, low tenure, and overall more entry level which leads us to segmenting people at risk in clusters 2 and 3.

demographic differences between employees who leave and those who remain?

There aren't many demographic differences from employees who leave vs those who remain. The demographic for males is anywhere from 54-67% so the gender gap isn't substantial. The strongest indicator isn't demographic differences but rather job tenure and seniority.

position-related factors, such as job level and department, associated with a higher probability of leaving IBM?

Yes, Department defines the four segments strongly, with job level tracking attrition closely. The average job level of 1.5 – 2.0 carries the highest attrition rate which tells us that a lower job level in the sales area carries the highest risk of exiting.

Does seniority, measured through total working experience, years at IBM, and years in the current role or with the current manager, play a role in an employee's decision to stay or leave?

Yes, these variables associated with seniority greatly affect employee tenure. Cluster 0 (senior) contains some of the highest paid, highest tenure, and job level out of all segments. The first few years of employment as a mid/manager role have the highest risk window.

Are job satisfaction and work-life balance associated with employee attrition?

Job satisfaction (2.62 – 2.78) and work life balance (2.71 – 2.92) are clear, but are a poor indicator of an exit risk signal given how nearly identical these scores are across each segment. For example, HR has the highest work life balance (2.92) but has an attrition rate of 20% way above the 16.2% average. Factors such as pay and tenure have a better segmentation signal.

Conclusion

The main cluster that needs to be addressed is the early career sales area and early careers in general. The two segments that need to be addressed are clusters 2 and 3. Early careers in sales has the highest attrition rate of 22% and R&D has an attrition rate of 17%. While 17% isn't the highest it has the greatest number of employees (716). Job level and tenure are the highest predictors of those at risk. While the exit interviews and frequent job surveys may help, they are not great at predicting exit risk. The scenario simulator built into the application will be a great tool to use when it comes to flagging employees at risk for exiting.